



Research ethics and issues regarding the use of ChatGPT-like artificial intelligence platforms by authors and reviewers: a narrative review

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Abstract

While generative artificial intelligence (AI) technology has become increasingly competitive since OpenAI introduced ChatGPT, its widespread use poses significant ethical challenges in research. Excessive reliance on tools like ChatGPT may intensify ethical concerns in scholarly articles. Therefore, this article aims to provide a comprehensive narrative review of the ethical issues associated with using AI in academic writing and to inform researchers of current trends. Our methodology involved a detailed examination of literature on ChatGPT and related research trends. We conducted searches in major databases to identify additional relevant articles and cited literature, from which we collected and analyzed papers. We identified major issues from the literature, categorized into problems faced by authors using nonacademic AI platforms in writing and challenges related to the detection and acceptance of AI-generated content by reviewers and editors. We explored eight specific ethical problems highlighted by authors and reviewers and conducted a thorough review of five key topics in research ethics. Given that non-academic AI platforms like ChatGPT often do not disclose their training data sources, there is a substantial risk of unattributed content and plagiarism. Therefore, researchers must verify the accuracy and authenticity of AI-generated content before incorporating it into their article, ensuring adherence to principles of research integrity and ethics, including avoidance of fabrication, falsification, and plagiarism.

Keywords

Artificial intelligence; Authorship; Peer review; Publishing; Research ethics

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Introduction

Background

The emergence of ChatGPT and fierce technological competition

ChatGPT, a generative artificial intelligence (AI) platform developed by OpenAI, was launched in December 2022 based on generative pretrained transformer 3.5 (GPT-3.5). It is reported that the early release of GPT-4, a multimodal GPT capable of responding to text questions with images, has reduced instances of hallucinations [1]. While ChatGPT is engineered to minimize false responses, it can still produce inaccurate, absurd, or unethical answers. OpenAI's recent development, Sora, a text-to-video model, generates videos that preserve the quality of the physical world and adhere to user prompts. These AIs, capable of generating sophisticated text, images, and videos, have attracted a large user base, highlighting the rapid advancements in AI technology. Consequently, a race among information companies to lead in generative AI has intensified. Examples include Google's Bard and its multimodal AI Gemini, Meta's LLaMA 2, Baidu's Ernie Bot, and Naver's HyperCLOVA X. Additionally, Coscientist, an AI dedicated to designing, planning, and conducting chemical experiments, has been developed [2]. The rapid expansion of AI technology has led the *Science* and *Nature* journals to recognize AI as one of the top achievements and contributors of 2023. AI technology is increasingly infiltrating everyday life, exemplified by Samsung's S24 mobile phone, which offers on-device AI interpretation and translation [3].

Changes in the research and publication landscape in response to ChatGPT

As the academic use of general-purpose generative AI has attracted attention, a review article has been published that examined the publication and research trends of ChatGPT-related articles from the perspective of their use, rather than their technical aspects [1]. That review identified a surge of publications, primarily consisting of rapid opinions and suggestions, during the initial phase of ChatGPT's introduction. One year later, the highest number of ChatGPT-related articles was recorded in Scopus, with PubMed in the medical field and Web of Science across all subjects showing similar figures. Despite a decrease in the proportion of articles in the medical field among ChatGPT-related articles with identifiable subject areas in major databases—from 71.4% of 935 articles in 7 months [1] to 55.7% of 3,222 articles in 12 months [3], the medical community's interest in this topic has persisted. The community is actively exploring the use of ChatGPT, although its practical application in medicine is still rare. Meanwhile, the education sector has raised numerous concerns about the educa-

tional implications, including the evaluation of assignments. The legal community has faced lawsuits concerning AI's intellectual ownership and unauthorized learning. Additionally, the journal publishing community has begun to establish early guidelines for AI usage in article writing [1]. Therefore, the necessity to clearly define the appropriate use of ChatGPT in scholarly writing has become evident. Generative AI's exceptional capability to assist with time-intensive tasks, such as reviewing prior research and analyzing data, has led to its increased adoption driven by the pressures of the publish-or-perish environment. The widespread use of ChatGPT-like AI platforms in scholarly articles has posed challenges to academic integrity, raising concerns about the reliability of peer review and the integrity of journal editing. Consequently, the excessive reliance on generative AI could further complicate issues related to research ethics in scholarly publications.

Appearance of academic AI trained on articles

Due to the impact of ChatGPT, as a nonacademic platform, Elsevier launched Scopus AI, which utilizes abstracts from articles published after 2013 in Scopus. This service, available since 2024, offers natural language summaries, visualized results, and suggested follow-up questions. Similarly, Elicit, which also relies on article abstracts, is accessible for a fee, while a private AI trial using Dimensions has been announced. Clarivate has initiated the development of an AI that is trained on Web of Science articles [3]. Unlike ChatGPT, which primarily sources its training data from publicly available internet content, these new types of academic generative AI, trained on bibliographic databases containing articles, aims to assist researchers in drafting introductions and literature reviews. Given that 62.2% of researchers consider the use of ChatGPT in article writing unethical [4], the ethical implications of AI-driven articles are a significant concern. Although academic AI requires either library subscriptions or personal expenses and has been slower to spread—having been introduced a year after ChatGPT—it represents a more controlled and potentially ethical approach. In contrast, nonacademic ChatGPT is both affordable and widely accessible.

Need for a review on the use of AI in article writing

Unlike academic AI, the use of general-purpose and nonacademic generative AI platforms such as ChatGPT raises several research ethics concerns. These include the quality of training data, bias in learning outcomes, the accuracy of generated content, and potential issues with unquoted material and plagiarism [5]. Therefore, researchers must strictly adhere to research ethics when using nonacademic AI tools to enhance their productivity in article writing. The ease of using AI tools like ChatGPT in articles, without proper citations and refer-

ences, could exacerbate research ethics issues due to fierce competition. While information technology (IT) tools such as Grammarly (Grammarly Inc) can improve the quality of writing without raising research ethics concerns, AI that instantly produces sophisticated text, images, and videos in response to user prompts presents various ethical issues in the learning and creation processes. According to a survey published in *Nature*, more than half of the 1,600 researchers surveyed expected AI tools to be very important or essential in their fields [6]. Nonetheless, about 68% believed these tools would make plagiarism easier, 55% were concerned that they could facilitate fraud, and 53% indicated that careless use might lead to irreproducible research. Consequently, there is an urgent need to provide researchers with clear guidelines on using AI for writing articles. These guidelines should cover the use of nonacademic generative AI, the detection and allowance of AI use during peer review, and the adherence to research ethics by both authors and reviewers.

Objectives

This article aims to comprehensively review the research ethics issues raised by using ChatGPT-like AIs in article writing and to share trends with researchers interested in utilizing AIs. For this purpose, we utilized the author's previous research on the academic usage trends of ChatGPT [1,3] as a foundation to identify additional papers on AI usage reported in major databases up to March 2024. We then conducted searches in PubMed for similar articles with citations in the literature and expanded our review through supplementary searches. The content of the final collection of articles was primarily examined for this purpose. The research ethics issues in articles using AI were categorized from the perspectives of authors, reviewers, and editors. These issues were thoroughly investigated and are discussed in this article from the standpoint of research ethics compliance. In this review, AI refers to nonacademically generated AI, such as ChatGPT, unless specified otherwise; it does not refer to academically generated AI, such as Scopus AI.

Ethics statement

This review was not a study with human subjects, so neither institutional review board approval nor informed consent was required.

Issues in Writing Articles Using AI That Authors Must Be Aware Of

What aspects of writing articles can be AI-assisted?

In writing articles, AI can be utilized for accurate translation, grammatical correctness, and idea generation, as well as for

summarizing content, and crafting conclusions [1]. Journal reviewers have found AI particularly useful for generating research ideas and summarizing data [7]. AI has also proven effective in generating ideas for systematic reviews in plastic surgery, addressing complex problems with accuracy and simplicity [8]. Surgical students in the United States have reported that they perceived articles generated by ChatGPT as more straightforward and better organized than those from evidence-based sources [9]. Although ChatGPT can produce more readable abstracts and respond concisely to critical information, the quality of these abstracts was generally rated as lower than that of the original texts [10]. A study comparing ChatGPT-generated abstracts with those from an obstetrics journal indicated that the median score—based on Grammarly scores, writing issues, and correctness errors—was lower than that of the original abstracts, although there were no grammatical inaccuracies [11]. In expert assessments of clinical reviews, AI-generated summaries were found to be more extractive and comprehensive, though human-generated summaries often included valuable interpretations that were missing from AI summaries [12]. ChatGPT has been noted for producing more refined sentences more quickly than traditional English proofreading services, making it a valuable tool for language editing [13]. AI offers numerous advantages throughout the various stages of article writing [14]. Upon reviewing the literature, the anticipated benefits of AI in article writing can be summarized as follows: researchers can use AI to discover, translate, and summarize articles and research trends, identify experimental methods and scientific knowledge, and compile results and statistics. For authors, AI can assist in drafting abstracts, keywords, introductions, literature reviews, data analyses, conclusions, and references, and it can help identify similarities for anti-AI checks. Reviewers can benefit from AI for similarity and anti-AI checks, locating peer reviewers, and drafting review results. Readers may find AI helpful for translating and summarizing content, discovering complementary knowledge, and organizing related information.

How many poor references and hallucinations are due to AI learning problems?

The evaluation scores assigned by reviewers to generated articles were ranked in the following order: abstract, literature review, research idea, methodology, citations, and references [15]. In that study, only 8% of the references were retrievable via Google Scholar and Mendeley. In other studies of article writing, the accuracy of references was low, with only 7% being correct [16] and 70% of the cited references being inaccurate [17], highlighting the need for careful reference checking when using AI in article writing. ChatGPT correctly answered questions from radiologists 67% of the time, and only 36.2%

of the references it provided contained accurate information [18]. In medical articles, 56% of references did not have a digital object identifier (DOI) or were not searchable on Google [19]. Many articles written by AI contained significant inaccuracies and fictitious references, potentially appearing credible to nonexpert readers [20]. The issue with such unreliable referencing is its plausibility, which can mislead uninformed authors or readers into accepting it as factual. Although AI-related errors and hallucinations are problematic, they are not quantifiable, making it challenging to locate prior research on these issues.

Can AI write both review articles and case reports?

The results of a trial on AI-only writing of scientific review articles were published in the journal *Current Osteoporosis Reports*. Human authors completed their tasks more quickly than AI, which required additional time for fact-checking and editing [21]. When AIs assist in review writing, they produce drafts more rapidly but necessitate extra time for content verification due to concerns about plagiarism and high similarity scores [17], or they require more editing because of hallucinations and plagiarism [22]. Similarly, while AI can independently write scientific review articles, the author must subsequently verify the scope and length of edits, the accuracy of claims, and the potential for plagiarism [23]. In a hybrid narrative review case involving collaboration between humans and ChatGPT, the results highlighted both the effectiveness and the concerns associated with ChatGPT [24]. Meanwhile, in evaluations of dermatology case reports by experts, AI-derived text received higher scores for quality and readability, yet human reviewers only correctly identified an AI-generated case report 35% of the time [25]. AI is also capable of simplifying text to a layperson's level, as demonstrated in the drafting of a fracture case report [26]. Although the case report had some errors, it was also accurate and complete [27]. However, a medical case report entirely generated by ChatGPT was found to have inadequate content, lacked references, and was deemed unsuitable for producing scientific articles [28]. Exploring ChatGPT's potential for creating clinical vignettes could lead to misinformation [29]. Ultimately, entrusting the complete creation of scientific reviews or case reports to AI is not advisable due to concerns over content accuracy and quality reliability. This necessitates additional time for authors to post-check and suggests that AI should be used primarily to design drafts or specific sections of these articles.

How often is AI used in articles?

In a ZeroGPT (ZeroGPT) test of each 200 *Orthopaedics & Traumatology: Surgery & Research* (OTSR) journal articles before and after the advent of ChatGPT, AI-generated text

was detected in 10.1% of abstracts and in 5.6% of the main texts [30]. Although the plagiarism rate remained unchanged, the incidence of AI-generated content was approximately 5% higher than previously observed, suggesting an association between the use of translation software and increased AI detection rates. A comparison of OTSR journal articles from 2022 and 2023 using ZeroGPT revealed that the average percentage of articles with suspected AI use was $11\% \pm 6\%$, indicating a rise in the utilization of AI [31]. An examination of acknowledgments that mentioned AI use in articles showed that medicine had the highest number of publications utilizing ChatGPT. Specifically, 24% of these articles acknowledged AI use in content generation, 37% recognized the contributions of OpenAI and ChatGPT, 41% mentioned their use for support and collaboration, and 53% cited AI use in their references [32].

Detection and Permissibility of AI Use That Reviewers Must Be Aware Of

Should AI authorship and attribution of AI be allowed?

After the release of ChatGPT, the journal publishing community quickly addressed the issue of AI authorship and the acceptability of AI-assisted articles [33]. The Committee on Publication Ethics (COPE) and the top five publishers [34]—Elsevier, Springer, MDPI, Wiley, and Taylor & Francis—took a similar stance, stating that if an AI platform is used, it is the responsibility of the author to acknowledge the use, but AI cannot be the author [35]. However, the journal *Science* has adopted a stricter policy, stating that using AI without permission is considered scientific misconduct [36]. The guidelines for AI use in most journals, except *Science*, are summarized as follows [1]: AI cannot be credited with authorship; human authors must assume full responsibility for the accuracy of the results; and there must be clear disclosure of the AI tools utilized. In 2021, the sharing rate of scientific articles from these top five publishers exceeded 62% [34]. Following the introduction of ChatGPT, many journals have adopted these AI authorship policies, which are now upheld by reviewers. Therefore, most articles, particularly those from the top five publishers, adhere to these policies. However, some journals still lack specific policies regarding AI authorship.

Can reviewers and editors determine whether an article is AI-assisted?

A previous study noted that human-written articles were specific, varied, and informative, whereas ChatGPT tended to use general terms, prioritizing fluency and logical coherence over contextual accuracy [37]. When orthopedic surgeons compared AI-generated abstracts to their original counterparts,

they found that the overall quality of human-written abstracts was superior, although the differences were not statistically significant [38]. In tests where human reviewers compared original and ChatGPT-generated abstracts, they correctly identified 68% of the ChatGPT abstracts but mistakenly labeled 14% of the original abstracts as AI-generated [39]. When only titles and abstracts were used for comparison, reviewers accurately identified 30% of ChatGPT abstracts and 43.7% of human abstracts, yet they struggled to distinguish between the two precisely [40]. Orthopedic residents and professors were able to correctly identify the authors of the abstracts 34.9% and 31.7% of the time, respectively, while two AI detectors achieved higher accuracy rates of 42.9% and 66.6% [41]. When reviewers were given AI-generated abstracts along with the full text of the journal, they achieved a 93% accuracy rate in identifying ChatGPT abstracts. The quality of these abstracts was significantly lower in unstructured formats, but no quality differences were observed in structured abstracts with four subheadings [42]. The quality of the generated abstracts did not meet the threshold for acceptance in peer-reviewed journals, and AI-generated text was more likely to be recognized by experienced reviewers [15]. Therefore, articles written by AI are often not detected by less experienced reviewers.

How accurate are tools for detecting AI-assisted articles?

Given the distinct characteristics of text written by humans and AI, algorithms can effectively detect AI-generated content [37]. AI writing in chemistry journals can be accurately detected, as demonstrated in article introductions [43]. Although this detector developed just 1 month after GPT-4 showed 100% accuracy in chemistry journals, its performance in other journals with a more liberal experimental process remains unknown. AI abstracts generated from the titles of medical journals were detected with over 99.9% accuracy by the GPT-2 Output Detector [39]. In a study, ZeroGPT identified AI-written texts with an accuracy of only about 35% to 65%, and OpenAI's text classifier achieved around 10% to 55% accuracy [44]. In evaluating the AI detection performance of tools like GPTZero (GPTZero Inc), ZeroGPT, Writer AI content detector (Writer Inc), and Originality (Originality.AI Inc), Originality excelled in distinguishing between AI-generated and human texts [45]. In one study, GPTZero showed a low classification rate for human text as machine-generated and a high rate for classifying machine-generated text as human [46]. However, concerns have been raised that paraphrasing may reduce the detection capabilities of GPT detectors [47]. Prevention is deemed more important than detection, as approximately 50% of AI-generated texts using obfuscation technology are misattributed to humans [48]. The accuracy rates were 96% for human-written texts, 76% for machine-translated to English, 74% for machine-

generated without modifications, 42% for machine-generated with subsequent human editing, and only 26% for machine-generated with subsequent machine paraphrasing [48]. Therefore, more accurate AI detection tools will be invaluable in peer review, content correction, and syntax improvement across various journals. These AI detection tools are not always correct or reliable, and content obfuscation techniques further degrade their performance. Testing the performance of AI detection tools with machine paraphraser, translators, and patch writers is essential. It is challenging to perfectly detect the behavior of paraphrasing text in an article, as current technology struggles to identify different expressions of the same meaning used to evade detection. This situation resembles the proverbial competition between spears and shields, reflecting the ongoing battle between generative AIs attempting to evade detection and machine tools striving to detect them more effectively. While article reviewers and editors rely on the support and robustness of these AI detection tools, their practical use is limited by technical challenges in detecting AI text that has been edited and paraphrased perfectly.

Should AI be used to support reviewers and editors in the review process?

When using ChatGPT, journal reviewers were only able to distinguish between AI-generated and human-written text 38.9% of the time, with 25.9% expressing disagreement, even though many editors considered AI acceptable for use [49]. Therefore, numerous journal editors deem AI-generated abstracts suitable for summarizing and editing text. AI detection tools used by reviewers struggle to identify AI-assisted content, underscoring the importance of weighing the risks and benefits of AI in the peer review process. In evaluations of MEDLINE abstracts, originality demonstrated 100% sensitivity, 95% specificity, and 97.6% accuracy in differentiating texts, although the likelihood of AI-generated text rose from 21.7% to 36.7% [50]. Quality assessments by human reviewers and ChatGPT for accepted articles were generally comparable. However, there were more pronounced differences in the scores for rejected articles, indicating less consistency in ChatGPT's evaluations [51]. Thus, AI can expedite the review process and support peer reviewers by enhancing efficiency. Nevertheless, the overarching conclusion from previous studies is that AI cannot fully replace human reviewers.

Research Ethics Using AI: Necessary Considerations for Authors, Reviewers, and Editors

The use of ChatGPT in article writing is likely to become more widespread, as it can be used for various purposes. Therefore, ChatGPT poses potential ethical challenges, including threats

AI use	Considerations when using generative AI in articles	
	Key issue	Research ethics
Authors	What aspects of writing articles can be AI-assisted? How many poor references and hallucinations are due to AI learning problems? Can AI write both review articles and case reports? How often is AI used in articles?	High similarity scores and copyright violation concerns in AI learning Concerns about low quality due to AI's poor citations and inadequate references Anxiety related to fabrication, falsification, plagiarism, and research integrity
Reviewers	Should AI authorship and attribution of AI be allowed? Can reviewers and editors determine whether an article is AI-assisted? How accurate are tools for detecting AI-assisted articles? Should AI be used to support reviewers and editors in the review process?	The need for uniform standards for AI attribution and inappropriate authorship Disrupting subsequent research with numerous retractions due to the overuse of AI

Fig. 1. The research framework of this review. AI, artificial intelligence.

to academic integrity and data accuracy [52]. To address these issues, a collaborative effort from all stakeholders in academia is essential. This includes improving peer review systems with technology designed to detect AI-generated plagiarism, advocating for higher ethical standards across the academic community, and providing clear guidelines for the responsible use of AI tools [53]. Therefore, the main ethical concerns and research issues that authors and reviewers should consider when incorporating AI into their articles are summarized in Fig. 1, focusing on representative issues.

High similarity scores and copyright violation concerns in AI learning

Generative AI primarily relies on publicly available data for learning, which means it may not access the most professional or up-to-date information. If inaccuracies from AI-generated content are incorporated into articles, issues such as data bias may arise. During its training, AI probabilistically generates text, often producing content that closely resembles existing research or sources. This similarity raises the potential for copyright infringement due to unattributed use of learning data and plagiarism. There have been instances of copyright litigation involving media data used in AI training, as well as the creation of new images from training images obtained without permission [1]. The European Commission has proposed a new legal framework for the intellectual property rights of AI-generated works, including a copyright exception for “text and data mining” [54]. Furthermore, in the United States, AI-generated works may qualify for copyright registration under certain conditions as outlined in the rulemaking guidelines. Following these developments, the European Parliament passed the Artificial Intelligence Act in March 2024. Consequently, if AI-generated content is used in articles without proper verification, the author could face copyright infringement charges, necessitating a thorough follow-up review.

Concerns about low quality due to AI’s poor citations and inadequate references

Ensuring the reliability and accuracy of article content is crucial for scientific research. Generating misleading or inaccurate content compromises scientific integrity. The excessive use of AI-generated content without proper citation can lead to inadvertent plagiarism. Plagiarism, whether it involves stealing words or ideas, can undermine research integrity [55]. While the plagiarism of ideas has traditionally been unacceptable, attitudes toward text plagiarism often vary depending on cultural perspectives [46]. Additionally, double plagiarism poses a significant issue when an author reuses plagiarized text without acknowledging the original source, which may have been learned by AI. However, it is challenging to determine definitively whether AI-generated content is original. Copyright infringements through plagiarism can lead to potential criminal consequences and civil liabilities. In the field of nephrology, ChatGPT has a citation accuracy rate of only 38% [56], which, although better than other AIs, still indicates a significant issue with referencing. Poor referencing by AIs can result in not only low-quality articles but also in the perpetuation of inaccuracies if authors do not verify the information. Nevertheless, we believe that the issues of missing citations and inadequate references in ChatGPT can be addressed by academic AIs like Scopus AI, which has been trained on scholarly articles.

Anxiety related to fabrication, falsification, plagiarism, and research integrity

Research integrity is fundamentally linked to the honesty and reliability of the research process. The use of AI to fabricate, falsify, or plagiarize content in research data or results represents a severe breach of this integrity [49]. Concerns arise when image or video-generating AI tools such as DALL-E 2 (OpenAI), Midjourney (Midjourney Inc), Stable Diffusion (Stability AI), or Sora are used to create images from tables

and figures for inclusion in articles without proper justification, leading to potential fabrication and falsification of research data. Additionally, if AI generates fabricated texts, transcripts of interviews, or responses to questions, it could lead to the spread of misinformation, dissemination of bias or discrimination, decreased transparency, privacy violations, and diminished researcher responsibility and accountability. The capability of reviewers and AI detection tools to identify AI-generated content is therefore a critical issue that journal publishers must urgently address [53]. As AI-generated content becomes more prevalent, it will be interesting to observe whether mandatory labeling of AI content effectively addresses issues of fabrication, falsification, and data security. Moreover, if AI randomly generates biased, discriminatory, and untruthful texts that are used in articles, issues of plagiarism and research integrity may emerge since these texts are characterized by falsehoods and distortions. Human authors must assume full responsibility for ensuring the originality, transparency, integrity, and validity of their AI-supported manuscripts. In the context of AI-assisted articles, we can anticipate the development of AI-supported reviewers that assist editors in ensuring the fairness and efficiency of the peer review process.

The need for uniform standards for AI attribution and inappropriate authorship

To date, most countries have maintained that for an AI creation to be eligible for copyright, it must reflect the thoughts and feelings of a human and exhibit a minimum level of creativity. If AI is incorrectly identified as the author, it is not recognized as such under copyright law due to its nonhuman status. This misattribution could also constitute a violation of copyright law. Although it is technically feasible to submit an article listing AI as an author, it is unlikely to be accepted. Many journals have policies that prevent AI from being credited as an author, insisting that the actual author must assume full responsibility for the content's integrity and adherence to research ethics [1]. An investigation of the AI authorship policies of 300 major journals found that 58.7% had an AI authorship policy, with 96.6% allowing AI to improve manuscript quality and 98.9% stating that AI cannot be an author [57]. However, a database search for articles that named ChatGPT as an author yielded 16 unique entries after one year [3]. These articles predominantly appear in journals, repositories, or archives that lack established AI authorship policies. Looking ahead, if the EU's proposal to grant copyrights to AI products is adopted, the way authors and papers utilizing AI products are cited and labeled might change. Nonetheless, there is a pressing need for the development of author policies and guidelines concerning AI use. These should be crafted in collaboration with the academic community to establish clear, common

standards for both authors and reviewers. Recommendations from research ethics committees could further guide these discussions on AI utilization [58].

Disrupting subsequent research with numerous retractions due to the overuse of AI

The use of AI to rapidly generate low-quality manuscripts and the proliferation of predatory journals that rely heavily on AI for rapid publication—as exemplified by the publication speed of MDPI [34]—pose significant concerns for publishers, editors, reviewers, readers, and other stakeholders in the publishing industry. This trend leads to wasted energy and time as these parties strive to verify the authenticity of the content. It is crucial to differentiate between human-generated and AI-assisted content, given that the imperfections in AI detection tools make accurate classification both time-consuming and expensive. If technological measures are not implemented early in the review process to assess AI-assisted text and images, there is a risk of producing spurious or low-quality articles, potentially leading to an increased rate of retractions.

Notably, many journals that were removed from the Web of Science list in 2023 belonged to Hindawi, a publisher acquired by Wiley. Along with removing these journals, the number of articles produced by “paper mills” was estimated to be in the hundreds of thousands [59]. In 2023, Wiley-Hindawi retracted over 8,000 articles from a total of approximately 10,000 due to issues such as incoherent text and irrelevant references. Many of these articles were from special issues overseen by guest editors. This led to Wiley's decision to discontinue the Hindawi brand name and a loss of US \$35 million in annual revenue [59]. When an article is retracted and removed from bibliographic databases due to suspicions of inadequate peer review or excessive AI assistance, the credibility of subsequent studies or reviews that cite or discuss it is compromised, thereby disrupting the knowledge ecosystem and societal trust. ChatGPT can significantly enhance research by aiding in idea generation, content creation, and experience direction, provided it is used within the guidelines and limitations set by journals. If properly implemented, ChatGPT can improve the quality of articles and research ethics. However, AI cannot completely replace the critical roles of human authors and reviewers [60].

Conclusion

With the emergence of ChatGPT, competition in the field of generative AI technology is growing, and the application of general-purpose, nonacademic generative AIs like ChatGPT in scholarly articles is becoming more common. Utilizing ChatGPT in scientific research poses potential risks to the in-

tegrity of research, sparking concerns about the accuracy and originality of published articles. In this study, we initially reviewed existing research to pinpoint issues that both authors and reviewers need to be aware of when incorporating AI in the processes of article writing and peer review. We then explored the ethical challenges faced by authors and reviewers using ChatGPT-like AIs, including issues such as high similarity in AI outputs, copyright law violations, compromised quality due to inadequate citations and unquoted sources, concerns over research integrity (including fabrication, falsification, and plagiarism), the necessity for standardized AI authorship guidelines, inappropriate attribution of authorship, and the potential disruption of future research due to numerous retractions [61–63] stemming from AI overuse.

While academic generative AIs are significantly less prone to hallucinations and illusions associated with poor referencing, nonacademic generative AIs such as ChatGPT are inherently more susceptible to unquote and plagiarism due to the nondisclosure of trained sources. This review reveals that in many articles, AI is not permitted to be listed as an author in principle, and human authors must assume responsibility for all fact-checking and research integrity. Additionally, the identification of AI-generated content remains imperfect for reviewers and AI detection tools, necessitating their use in the submission and review process. It is also essential to differentiate between the legal issues in the learning process and research ethics when utilizing content. Moreover, a unified AI-related policy could provide guidance and transparency for authors, readers, reviewers, editors, and publishers. Addressing the research ethics issues associated with AI use will require education on writing articles with research integrity, which prevents fabrication, falsification, and plagiarism. Consequently, ChatGPT should be considered an additional AI tool to expedite the production of high-quality articles. However, this review is based on the assumption that all results from previous studies are valid, and the accuracy and truthfulness of this review are limited to the reliability of these cited studies, reflecting the author's lack of expertise in all of the discussed domains.

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References

1. Kim SJ. Trends in research on ChatGPT and adoption-related issues discussed in articles: a narrative review. *Sci Ed* 2024;11:3–11. <https://doi.org/10.6087/kcse.321>
2. Boiko DA, MacKnight R, Kline B, Gomes G. Autonomous chemical research with large language models. *Nature* 2023; 624:570–8. <https://doi.org/10.1038/s41586-023-06792-0>
3. Kim SJ. Research trends in ChatGPT-related articles and implications from performance evaluation of ChatGPT. *KSLA Insight* 2024;1:1–18. <https://doi.org/10.23234/KSLAIN.PUB.1.1.1>
4. Eppler M, Ganjavi C, Ramacciotti LS, et al. Awareness and use of ChatGPT and large language models: a prospective cross-sectional global survey in urology. *Eur Urol* 2024;85: 146–53. <https://doi.org/10.1016/j.eururo.2023.10.014>
5. Rao KN, Arora RD, Dange P, Nagarkar NM. NLP AI models for optimizing medical research: demystifying the concerns. *Indian J Surg Oncol* 2023;14:854–8. <https://doi.org/10.1007/s13193-023-01791-z>
6. Van Noorden R, Perkel JM. AI and science: what 1,600 researchers think. *Nature* 2023;621:672–5. <https://doi.org/10.1038/d41586-023-02980-0>
7. Dowling M, Lucey B. ChatGPT for (finance) research: the Bananarama Conjecture. *Financ Res Lett* 2023;53:103662. <https://doi.org/10.1016/j.frl.2023.103662>
8. Gupta R, Herzog I, Weisberger J, Chao J, Chaiyasate K, Lee ES. Utilization of ChatGPT for plastic surgery research: friend or foe? *J Plast Reconstr Aesthet Surg* 2023;80:145–7. <https://doi.org/10.1016/j.bjps.2023.03.004>
9. Breeding T, Martinez B, Patel H, et al. The utilization of ChatGPT in reshaping future medical education and learning perspectives: a curse or a blessing? *Am Surg* 2024;90: 560–6. <https://doi.org/10.1177/00031348231180950>
10. Hwang T, Aggarwal N, Khan PZ, et al. Can ChatGPT assist authors with abstract writing in medical journals? Evaluating the quality of scientific abstracts generated by ChatGPT and original abstracts. *PLoS One* 2024;19:e0297701. <https://doi.org/10.1371/journal.pone.0297701>
11. Levin G, Meyer R, Kadoch E, Brezinov Y. Identifying

- ChatGPT-written OBGYN abstracts using a simple tool. *Am J Obstet Gynecol MFM* 2023;5:100936. <https://doi.org/10.1016/j.ajogmf.2023.100936>
12. Tang L, Sun Z, Idnay B, et al. Evaluating large language models on medical evidence summarization. *NPJ Digit Med* 2023;6:158. <https://doi.org/10.1038/s41746-023-00896-7>
13. Kim SG. Using ChatGPT for language editing in scientific articles. *Maxillofac Plast Reconstr Surg* 2023;45:13. <https://doi.org/10.1186/s40902-023-00381-x>
14. He S, Yang F, Zuo JP, Lin ZM. ChatGPT for scientific paper writing: promises and perils. *Innovation (Camb)* 2023; 4:100524. <https://doi.org/10.1016/j.xinn.2023.100524>
15. Khlaif ZN, Mousa A, Hattab MK, et al. The potential and concerns of using AI in scientific research: ChatGPT performance evaluation. *JMIR Med Educ* 2023;9:e47049. <https://doi.org/10.2196/47049>
16. Bhattacharyya M, Miller VM, Bhattacharyya D, Miller LE. High rates of fabricated and inaccurate references in ChatGPT-generated medical content. *Cureus* 2023;15:e39238. <https://doi.org/10.7759/cureus.39238>
17. Kacena MA, Plotkin LI, Fehrenbacher JC. The use of artificial intelligence in writing scientific review articles. *Curr Osteoporos Rep* 2024;22:115–21. <https://doi.org/10.1007/s11914-023-00852-0>
18. Wagner MW, Ertl-Wagner BB. Accuracy of information and references using ChatGPT-3 for retrieval of clinical radiological information. *Can Assoc Radiol J* 2024;75:69–73. <https://doi.org/10.1177/08465371231171125>
19. Athaluri SA, Manthena SV, Kesapragada VS, Yarlagadda V, Dave T, Duddumpudi RT. Exploring the boundaries of reality: investigating the phenomenon of artificial intelligence hallucination in scientific writing through ChatGPT references. *Cureus* 2023;15:e37432. <https://doi.org/10.7759/cureus.37432>
20. Ariyaratne S, Iyengar KP, Nischal N, Chitti Babu N, Botchu R. A comparison of ChatGPT-generated articles with human-written articles. *Skeletal Radiol* 2023;52:1755–8. <https://doi.org/10.1007/s00256-023-04340-5>
21. Awosanya OD, Harris A, Creecy A, et al. The utility of AI in writing a scientific review article on the impacts of COVID-19 on musculoskeletal health. *Curr Osteoporos Rep* 2024;22:146–51. <https://doi.org/10.1007/s11914-023-00855-x>
22. Margetts TJ, Karnik SJ, Wang HS, et al. Use of AI language engine ChatGPT 4.0 to write a scientific review article examining the intersection of Alzheimer's disease and bone. *Curr Osteoporos Rep* 2024;22:177–81. <https://doi.org/10.1007/s11914-023-00853-z>
23. Nazzal MK, Morris AJ, Parker RS, et al. Using AI to write a review article examining the role of the nervous system on skeletal homeostasis and fracture healing. *Curr Osteoporos Rep* 2024;22:217–21. <https://doi.org/10.1007/s11914-023-00854-y>
24. Temsah O, Khan SA, Chaiah Y, et al. Overview of early ChatGPT's presence in medical literature: insights from a hybrid literature review by ChatGPT and human experts. *Cureus* 2023;15:e37281. <https://doi.org/10.7759/cureus.37281>
25. Dunn C, Hunter J, Steffes W, et al. Artificial intelligence-derived dermatology case reports are indistinguishable from those written by humans: a single-blinded observer study. *J Am Acad Dermatol* 2023;89:388–90. <https://doi.org/10.1016/j.jaad.2023.04.005>
26. Bosbach WA, Senge JF, Nemeth B, et al. Ability of ChatGPT to generate competent radiology reports for distal radius fracture by use of RSNA template items and integrated AO classifier. *Curr Probl Diagn Radiol* 2024;53:102–10. <https://doi.org/10.1067/j.cpradiol.2023.04.001>
27. Jeblick K, Schachtner B, Dext J, et al. ChatGPT makes medicine easy to swallow: an exploratory case study on simplified radiology reports. *Eur Radiol* 2024;34:2817–25. <https://doi.org/10.1007/s00330-023-10213-1>
28. Buholayka M, Zouabi R, Tadinada A. Is ChatGPT ready to write scientific case reports independently? A comparative evaluation between human and artificial intelligence. *Cureus* 2023;15:e39386. <https://doi.org/10.7759/cureus.39386>
29. Le DP, Hall SC. Medical literature writing with ChatGPT: a rare case of choriocarcinoma syndrome with hemorrhagic brain metastases due to burned out metastatic mixed testicular cancer. *Cureus* 2023;15:e36655. <https://doi.org/10.7759/cureus.36655>
30. Bisi T, Risser A, Clavert P, Migaud H, Dartus J. What is the rate of text generated by artificial intelligence over a year of publication in Orthopedics & Traumatology: Surgery & Research? Analysis of 425 articles before versus after the launch of ChatGPT in November 2022. *Orthop Traumatol Surg Res* 2023;109:103694. <https://doi.org/10.1016/j.otsr.2023.103694>
31. Maroteau G, An JS, Murgier J, Hulet C, Ollivier M, Ferreira A. Evaluation of the impact of large language learning models on articles submitted to Orthopaedics & Traumatology: Surgery & Research (OTSR): a significant increase in the use of artificial intelligence in 2023. *Orthop Traumatol Surg Res* 2023;109:103720. <https://doi.org/10.1016/j.otsr.2023.103720>
32. Raman R. Transparency in research: an analysis of ChatGPT usage acknowledgment by authors across disciplines and geographies. *Account Res* 2023 Oct 25 [Epub]. <https://doi.org/10.1080/08989621.2023.2273377>
33. Thorp HH. ChatGPT is fun, but not an author. *Science*

- 2023;379:313. <http://doi.org/10.1126/science.adg7879>
34. Kim SJ, Park KS. Publishing trends of journals and articles in Journal Citation Reports during the COVID-19 pandemic: a descriptive study. *Sci Ed* 2023;10:78–86. <https://doi.org/10.6087/kcse.300>
 35. Lo CK. What is the impact of ChatGPT on education? A rapid review of the literature. *Educ Sci* 2023;13:410. <https://doi.org/10.3390/educsci13040410>
 36. Stokel-Walker C, Van Noorden R. What ChatGPT and generative AI mean for science. *Nature* 2023;614:214–6. <https://doi.org/10.1038/d41586-023-00340-6>
 37. Liao W, Liu Z, Dai H, et al. Differentiating ChatGPT-generated and human-written medical texts: quantitative study. *JMIR Med Educ* 2023;9:e48904. <https://doi.org/10.2196/48904>
 38. Lawrence KW, Habibi AA, Ward SA, Lajam CM, Schwarzkopf R, Rozell JC. Human versus artificial intelligence-generated arthroplasty literature: a single-blinded analysis of perceived communication, quality, and authorship source. *Int J Med Robot* 2024;20:e2621. <https://doi.org/10.1002/rcs.2621>
 39. Gao CA, Howard FM, Markov NS, et al. Comparing scientific abstracts generated by ChatGPT to real abstracts with detectors and blinded human reviewers. *NPJ Digit Med* 2023;6:75. <https://doi.org/10.1038/s41746-023-00819-6>
 40. Levin G, Meyer R, Yasmeen A, et al. Chat Generative Pre-trained Transformer-written obstetrics and gynecology abstracts fool practitioners. *Am J Obstet Gynecol MFM* 2023; 5:100993. <https://doi.org/10.1016/j.ajogmf.2023.100993>
 41. Makiev KG, Asimakidou M, Vasios IS, et al. A study on distinguishing ChatGPT-generated and human-written orthopaedic abstracts by reviewers: decoding the discrepancies. *Cureus* 2023;15:e49166. <https://doi.org/10.7759/cureus.49166>
 42. Cheng SL, Tsai SJ, Bai YM, et al. Comparisons of quality, correctness, and similarity between ChatGPT-generated and human-written abstracts for basic research: cross-sectional study. *J Med Internet Res* 2023;25:e51229. <https://doi.org/10.2196/51229>
 43. Desaire H, Chua AE, Kim MG, Hua D. Accurately detecting AI text when ChatGPT is told to write like a chemist. *Cell Rep Phys Sci* 2023;4:101672. <https://doi.org/10.1016/j.xcrp.2023.101672>
 44. Prillaman M. ‘ChatGPT detector’ catches AI-generated papers with unprecedented accuracy. *Nature* 2023 Nov 6 [Epub]. <https://doi.org/10.1038/d41586-023-03479-4>
 45. Bellini V, Semeraro F, Montomoli J, Cascella M, Bignami E. Between human and AI: assessing the reliability of AI text detection tools. *Curr Med Res Opin* 2024;40:353–8. <https://doi.org/10.1080/03007995.2024.2310086>
 46. Habibzadeh F. GPTZero performance in identifying artificial intelligence-generated medical texts: a preliminary study. *J Korean Med Sci* 2023;38:e319. <https://doi.org/10.3346/jkms.2023.38.e319>
 47. Anderson N, Belavy DL, Perle SM, et al. AI did not write this manuscript, or did it? Can we trick the AI text detector into generated texts? The potential future of ChatGPT and AI in Sports & Exercise Medicine manuscript generation. *BMJ Open Sport Exerc Med* 2023;9:e001568. <https://doi.org/10.1136/bmjsem-2023-001568>
 48. Weber-Wulff D, Anohina-Naumeca A, Bjelobaba S, et al. Testing of detection tools for AI-generated text. *Int J Educ Integr* 2023;19:26. <https://doi.org/10.1007/s40979-023-00146-z>
 49. Casal JE, Kessler M. Can linguists distinguish between Chat GPT/AI and human writing?: a study of research ethics and academic publishing. *Res Methods Appl Linguist* 2023; 2:100068. <https://doi.org/10.1016/j.rmal.2023.100068>
 50. Miller LE, Bhattacharyya D, Miller VM, Bhattacharyya M. Recent trend in artificial intelligence-assisted biomedical publishing: a quantitative bibliometric analysis. *Cureus* 2023;15:e39224. <https://doi.org/10.7759/cureus.39224>
 51. Saad A, Jenko N, Ariyaratne S, et al. Exploring the potential of ChatGPT in the peer review process: an observational study. *Diabetes Metab Syndr* 2024;18:102946. <https://doi.org/10.1016/j.dsx.2024.102946>
 52. Xu X, Chen Y, Miao J. Opportunities, challenges, and future directions of large language models, including ChatGPT in medical education: a systematic scoping review. *J Educ Eval Health Prof* 2024;21:6. <https://doi.org/10.3352/jeehp.2024.21.6>
 53. Miao J, Thongprayoon C, Suppadungsuk S, Garcia Valencia OA, Qureshi F, Cheungpasitporn W. Ethical dilemmas in using AI for academic writing and an example framework for peer review in nephrology academia: a narrative review. *Clin Pract* 2023;14:89–105. <https://doi.org/10.3390/clinpract14010008>
 54. Zhuk A. Navigating the legal landscape of AI copyright: a comparative analysis of EU, US, and Chinese approaches. *AI Ethics* 2023 May 30 [Epub]. <https://doi.org/10.1007/s43681-023-00299-0>
 55. Hosseini M, Rasmussen LM, Resnik DB. Using AI to write scholarly publications. *Account Res* 2023 Jan 25 [Epub]. <https://doi.org/10.1080/08989621.2023.2168535>
 56. Aiumtrakul N, Thongprayoon C, Suppadungsuk S, et al. Navigating the landscape of personalized medicine: the relevance of ChatGPT, BingChat, and Bard AI in nephrology literature searches. *J Pers Med* 2023;13:1457. <https://doi.org/10.3390/jpm13101457>
 57. Lund BD, Naheem KT. Can ChatGPT be an author? A

- study of artificial intelligence authorship policies in top academic journals. *Learn Publ* 2024;37:13–21. <https://doi.org/10.1002/leap.1582>
58. Pournaras E. Science in the era of ChatGPT, large language models and generative AI: challenges for research ethics and how to respond [Preprint]. *arXiv*; [posted 2023 May 24; updated 2023 Jul 29]. <https://doi.org/10.48550/arXiv.2305.15299>
59. Van Noorden R. More than 10,000 research papers were retracted in 2023: a new record. *Nature* 2023;624:479–81. <https://doi.org/10.1038/d41586-023-03974-8>
60. Benichou L; ChatGPT. The role of using ChatGPT AI in writing medical scientific articles. *J Stomatol Oral Maxillofac Surg* 2023;124:101456. <https://doi.org/10.1016/j.jormas.2023.101456>
61. Frontiers Editorial Office. Retraction: cellular functions of spermatogonial stem cells in relation to JAK/STAT signaling pathway. *Front Cell Dev Biol* 2024;12:1386861. <https://doi.org/10.3389/fcell.2024.1386861>
62. Zhang M, Wu L, Yang T, Zhu B, Liu Y. Retraction notice to “The three-dimensional porous mesh structure of Cu-based metal-organic-framework: aramid cellulose separator enhances the electrochemical performance of lithium metal anode batteries” [Surfaces and Interfaces, Volume 46, March 2024, 104081]. *Surf Interfaces* 2024;46:104550. <https://doi.org/10.1016/j.surfin.2024.104550>
63. Bader R, Imam A, Alnees M, et al. Removal notice to “Successful management of an Iatrogenic portal vein and hepatic artery injury in a 4-month-old female patient: a case report and literature review” [Radiology Case Reports 19 (2024) 2106–2111]. *Radiol Case Rep* 2024;19:3598. <https://doi.org/10.1016/j.radcr.2024.05.010>